

Ruben New Potentials Notes

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Abstract

As now we have lattice data for the potentials Σ_g^+ , Π_u and Σ_u^- we can change them in the calculation program and recompute the spectrum and also the mixing and the hyperfine splitting.

1 Static Potentials

For all three potentials we are going to do an interpolation between their behaviour at short- and at long-distances.

1.1 Short distances

For short distances we can use perturbative theory from pNRQCD. From [4], we get the perturbative $V^{(0)}$ up to $\mathcal{O}(\alpha_s^4)$ including the leading ultrasoft corrections $\delta a_3^{us}[\nu, \nu_{us}]$, and the leading renormalon ambiguity ($C_f = (N_c^2 - 1)/(2N_c)$, $C_A = N_c$)

$$V^{(0)} = -\frac{C_f \alpha_s}{r} \left[1 + \sum_{n=1}^3 \left(\frac{\alpha_s}{4\pi} \right)^n a_n(\nu, \nu_{us}; r) \right] + 2\delta m_{RS'}(\nu_f), \quad (1)$$

up to this order the coefficients a_n are completely known:

$$\begin{aligned} a_1(\nu; r) &= a_1 + 2\beta_0 \ln(\nu e^{\gamma_E} r), \\ a_2(\nu; r) &= a_2 + \frac{\pi^2}{3} \beta_0^2 + (4a_1\beta_0 + 2\beta_1) \ln(\nu e^{\gamma_E} r) + 4\beta_0^2 \ln^2(\nu e^{\gamma_E} r), \\ a_3(\nu; r) &= a_3 + a_1\beta_0^2\pi^2 + \frac{5\pi^2}{6} \beta_0\beta_1 + \textcolor{red}{16}\zeta_3\beta_0^3 \\ &\quad + \left(2\pi^2\beta_0^3 + 6a_2\beta_0 + 4a_1\beta_1 + 2\beta_2 + \frac{16}{3}C_A^3\pi^2 \right) \ln(\nu e^{\gamma_E} r) \\ &\quad + \left(12a_1\beta_0^2 + 10\beta_0\beta_1 \right) \ln^2(\nu e^{\gamma_E} r) + 8\beta_0^3 \ln^3(\nu e^{\gamma_E} r) \\ &\quad + \delta a_3^{us}(\nu, \nu_{us}). \end{aligned} \quad (2)$$

The factor 16 is 13 in Ruben's code and mine, should be checked. For the ultrasoft corrections to the static potential we take

$$\delta a_3^{us}(\nu, \nu_{us}) = \frac{16}{3} C_A^3 \pi^2 \ln\left(\frac{\nu_{us}}{\nu}\right). \quad (3)$$

The $\delta m_{RS'}(\nu_f)$ term appears to correct the shifting due to renormalons and can be expressed as a series in different orders in perturbation theory. It depends on 0, 1, or 2 loop terms around the renormalon and it reads as

$$\delta m_{RS'}(\nu, \nu_f) = \nu_f \left[\alpha_S(\nu) \delta m^{(0)} + \alpha_S^2(\nu) \delta m^{(1)} + \alpha_S^3(\nu) \delta m^{(2)} + \alpha_S^4(\nu) \delta m^{(3)} \right] \quad (4)$$

Having from [4] $\delta m^{(0)} = 0$ and $d_n(\nu, \nu_f) = \beta_n/2^{1+2n} \ln\left(\frac{\nu}{\nu_f}\right)$

$$\begin{aligned}\delta m^{(1)}\left(\frac{\nu_f}{\nu}\right) &= N_m \frac{\beta_0}{2\pi} S(1, b), \\ \delta m^{(2)}\left(\frac{\nu_f}{\nu}\right) &= N_m \left(\frac{\beta_0}{2\pi}\right) \left[S(1, b) \frac{2d_0(\nu, \nu_f)}{\pi} + \left(\frac{\beta_0}{2\pi}\right) S(2, b) \right], \\ \delta m^{(3)}\left(\frac{\nu_f}{\nu}\right) &= N_m \left(\frac{\beta_0}{2\pi}\right) \times \\ &\quad \times \left[S(1, b) \frac{3d_0^2(\nu, \nu_f) + 2d_1(\nu, \nu_f)}{\pi^2} + \left(\frac{\beta_0}{2\pi}\right) S(2, b) \frac{3d_0(\nu, \nu_f)}{\pi} + \left(\frac{\beta_0}{2\pi}\right)^2 S(3, b) \right].\end{aligned}\tag{5}$$

where

$$S(n, b) = \sum_{k=0}^2 c_k \frac{\Gamma(n+1+b-k)}{\Gamma(1+b-k)}\tag{6}$$

with $c_0 = 1$ and

$$b = \frac{\beta_1}{2\beta_0^2}, \quad c_1 = \frac{1}{4b\beta_0^3} \left(\frac{\beta_1^2}{\beta_0} - \beta_2 \right)\tag{7}$$

and

$$c_2 = \frac{1}{b(b-1)} \frac{\beta_1^4 + 4\beta_0^3\beta_1\beta_2 - 2\beta_0\beta_1^2\beta_2 + \beta_0^2(-2\beta_1^3 + \beta_2^2) - 2\beta_0^4\beta_3}{32\beta_0^8}.\tag{8}$$

With this our **perturbative** potentials read

$$V_{\text{Pert}}^{\Sigma_g} = V^{(0)}, \quad V_{\text{Pert}}^{\Pi_u} = V_{\text{Pert}}^{\Sigma_u^-} = -V^{(0)}/8.\tag{9}$$

We can see that $V_{\text{Pert}}^{\Sigma_g}$ is directly the singlet potential, the other two are the octet perturbative potentials so to write as the singlet we need to scale it using the colour factor $(C_A/2 - C_F)/(-C_F) = -1/8$. Comparing from short distance lattice data from ref. [5] we see that to reproduce correctly the data we need to add a global shift. So the **short distance** potentials finally become

$$V_{\text{Short}}^{\Lambda_\eta^\sigma} = V_{\text{Pert}}^{\Lambda_\eta^\sigma} + \mu_s^{\Lambda_\eta^\sigma}.\tag{10}$$

1.2 Long distances

For the long distance we use the expression of EST with the hole contributions.

$$V_{\text{EST}}(N) = \sqrt{\sigma^2 r^2 + 2\pi \left(N - \frac{1}{12}\right) \sigma}\tag{11}$$

where $\sigma = 0.21 \text{ GeV}^2$ is the string tension. N is the excitation levels of the EST potentials [2]. For the ground state ($N = 0, \Sigma_g$) and the excited strings ($N = 1, \Pi_u$ and $N = 3, \Sigma_u^-$). We use the expansions truncated up to $\mathcal{O}(1/r^3)$ for $N = 0$, while for the excited states using the full expression shows better convergence.

Once again to fit the lattice data we add a shift this time

$$V_{\text{Long}}^{\Lambda_\eta^\sigma} = V_{\text{EST}}^{\Lambda_\eta^\sigma} + \mu_l^{\Lambda_\eta^\sigma}.\tag{12}$$

V	x	y _f	n	m	r _c	$\mu_s - \mu_l$	a _m	a _{m+1}	a _{m+2}	b _m	b _{m+1}
SigG	1	1	9	3	0.66	0.73	-0.74	0.00	1.01	0.00	0.00
PiU	1	1	9	3	1.56	1.12	3.78	0.00	1.31	2.09	1.99
SigU	1	1	9	3	3.13	1.12	38.49	150.01	103.75	19.61	171.22

Figure 1: Fitting coefficients with first correction $m = 3$, last correction $k_a = 5$ ($k_b = 4$ for the denominator), and spacing $n = 9$. We extract $\mu_s - \mu_l$ by adjusting the short and long distance behaviour for short and long Lattice data independently.

1.3 Interpolation

The interpolation is done in a very complicated form and the coefficients are exactly adjusted to fit the lattice data the most accurately.

$$V_{\text{Fit}}^{\Lambda_\eta^\sigma}(r) = \frac{V_{\text{Short}}(r) + \Lambda_{\text{lat}} \sum_{j=m}^{k_a} a_j (r/r_c)^j + (r/r_c)^n V_{\text{Long}}(r)}{1 + \sum_{j=m}^{k_b} b_j (r/r_c)^j + (r/r_c)^n}, \quad (13)$$

the fitted data being: $n = 9$, $m = 3$, $k_a = 5$, $k_b = 4$; and r_c , $\mu_s - \mu_l$, a_j , b_j which depend on the potential Λ_η^σ in Fig. 1.

2 Static potential matrices

- **Quarkonium spin 0:** the shape of the potential matrix/hamiltonian is 1-dim for every $L = 0, 1, 2$.

$$\frac{\mathcal{J}(\mathcal{J} + 1)}{r^2} + mV_{\Sigma_g^+}(r) \quad (14)$$

- **Quarkonium spin 1:** is degenerate with quarkonium spin 0 so the spectrum is the same. This degeneracy is broken by the mixing in next section. Also it could be broken by the hyperfine but in quarkonium this is a NNLO effect.
- **Hybrid spin 0:** the shape of the potential matrices is the one from equations (6) and (7) of [3]. In here I flipped the order of the multiplet so our wave function will be

$$\begin{pmatrix} P_J^0(r) \\ P_J^+(r) \\ P_J^-(r) \end{pmatrix}.$$

As $S = 0 \rightarrow J = \mathcal{J}$. $J = 0$ is a special one-dimensional case, for $J \neq 0$ the matrix is three-dimensional. With the mixing we have three more states ordered as in Eq. (15) and (16).

- **Hybrids spin 1:** as before at the order of the static potentials this spectrum is degenerate with the hybrid spin 0 one. After including the mixing we will break this degeneracy that will be broken further by the inclusion of the hyperfine splitting.

3 Mixing potentials and matrices

Once understood this part which is maybe the most theoretical, the mixing potentials are also given by Ruben with a similar Table. This mixing potentials V_S^Σ and V_S^Π appear in the mixing matrices below. These matrices also contain the regular spectrum potentials, so they are the addition of the last section (regular spectrum) with this one (mixing). All the potentials here are defined by Ruben in his notes.

A very important point here is that in my MATLAB code the implementation of these matrices is done multiplying by m because of the way the coupled-channel Schrödinger equation solving script is constructed. So, everything is multiplied by m .

3.1 Mixing between Quarkonium spin 1 and Hybrid spin 0

When the hybrid $J = \mathcal{J}$ and the quarkonium $L = \mathcal{J}$

$$m_Q \left[-\frac{1}{m_Q} \frac{\partial^2}{\partial r^2} + \begin{pmatrix} \frac{\mathcal{J}(\mathcal{J}+1)}{m_Q r^2} + V_{\Sigma_g^+} & 2V_S^\Pi \\ 2V_S^\Pi & \frac{\mathcal{J}(\mathcal{J}+1)}{m_Q r^2} + V_{\Pi_u} \end{pmatrix} - E \right] \begin{pmatrix} R_{1\mathcal{J}\mathcal{M}}^0 \\ P_{0\mathcal{J}\mathcal{M}}^0 \end{pmatrix} = 0. \quad (15)$$

$2V_S^\Pi$ is $2V_S^\Sigma$ in Ruben's code. I used Π in mine. For $\mathcal{J} = 0$ the equation do not exist: since the gluon spin is 1 for hybrids ($J = L \oplus 1$), then only $L = 1$ can give $J = 0$, which is incompatible with the requirement of the previous equation $\mathcal{J} = J = L$.

When the hybrid $J = \mathcal{J}$ and the quarkonium $L = J \pm 1$

$$m_Q \left(\begin{array}{cc|cc} \frac{(\mathcal{J}+1)(\mathcal{J}+2)}{m_Q r^2} + V_{\Sigma_g^+} & & 2(V_S^\Pi + \frac{\mathcal{J}+1}{2\mathcal{J}+1} V_S^q) & -2V_S^q \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} \\ & \frac{(\mathcal{J}-1)\mathcal{J}}{m_Q r^2} + V_{\Sigma_g^+} & -2V_S^q \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} & 2(V_S^\Pi + \frac{\mathcal{J}}{2\mathcal{J}+1} V_S^q) \\ \hline 2(V_S^\Pi + \frac{\mathcal{J}+1}{2\mathcal{J}+1} V_S^q) & -2V_S^q \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} & \frac{(\mathcal{J}+1)(\mathcal{J}+2)}{m_Q r^2} + V_{\Sigma_u^-} + \frac{\mathcal{J}}{2\mathcal{J}+1} V_q & \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} V_q \\ -2V_S^q \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} & 2(V_S^\Pi + \frac{\mathcal{J}}{2\mathcal{J}+1} V_S^q) & \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} V_q & \frac{(\mathcal{J}-1)\mathcal{J}}{m_Q r^2} + V_{\Sigma_u^-} + \frac{\mathcal{J}+1}{2\mathcal{J}+1} V_q \end{array} \right) \begin{pmatrix} R_{1\mathcal{J}\mathcal{M}}^+ \\ R_{1\mathcal{J}\mathcal{M}}^- \\ P_{0\mathcal{J}\mathcal{M}}^+ \\ P_{0\mathcal{J}\mathcal{M}}^- \end{pmatrix} \quad (16)$$

For $\mathcal{J} = 0$, the states with $L = \mathcal{J} - 1$ do not exist and Eq. (16) reduce to two coupled equations with only R_{100}^+ and P_{000}^+ .

3.2 Mixing between Quarkonium spin 0 and Hybrid spin 1

$P_{1\mathcal{J}\mathcal{M}}^{0+}$, $P_{1\mathcal{J}\mathcal{M}}^{0-}$, $P_{1\mathcal{J}\mathcal{M}}^{+0}$ and $P_{1\mathcal{J}\mathcal{M}}^{-0}$ do not couple to heavy quarkonium. The remaining do it according to the following equations. For $\mathcal{J} > 1$, with $L = J, J \pm 1, J = \mathcal{J}, \mathcal{J} \pm 1$:

$$m_Q \left(\begin{array}{c|cccccc} \frac{\mathcal{J}(\mathcal{J}+1)}{m_Q r^2} + V_{\Sigma_g^+} & V_S^{++} & V_S^{-+} & V_S^{+-} & V_S^{--} & 2V_S^\Pi \\ \hline V_S^{++} & V^{++} & V_q \frac{\sqrt{(\mathcal{J}+1)(\mathcal{J}+2)}}{2\mathcal{J}+3} & & & \\ V_S^{-+} & V_q \frac{\sqrt{(\mathcal{J}+1)(\mathcal{J}+2)}}{2\mathcal{J}+3} & V^{-+} & & & \\ \hline V_S^{+-} & & & V^{+-} & V_q \frac{\sqrt{(\mathcal{J}-1)\mathcal{J}}}{2\mathcal{J}-1} & \\ V_S^{--} & & & V_q \frac{\sqrt{(\mathcal{J}-1)\mathcal{J}}}{2\mathcal{J}-1} & V^{--} & \\ 2V_S^\Pi & & & & & V^{00} \end{array} \right) \begin{pmatrix} R_{0\mathcal{J}\mathcal{M}} \\ P_{1\mathcal{J}\mathcal{M}}^{++} \\ P_{1\mathcal{J}\mathcal{M}}^{+-} \\ P_{1\mathcal{J}\mathcal{M}}^{+0} \\ P_{1\mathcal{J}\mathcal{M}}^{-0} \\ P_{1\mathcal{J}\mathcal{M}}^{00} \end{pmatrix} \quad (17)$$

For $\mathcal{J} = 0$, P_{100}^{00} , P_{100}^{--} , and P_{100}^{+-} do not exist and the system reduces to the three upper equations. For $\mathcal{J} = 1$, $P_{11\mathcal{M}}^{--}$ does not exist and the system above reduces to five coupled equations.

4 Hyperfine potentials and matrices

The hyperfine potentials are to be taken into account for hybrids of spin 1.

In here an important point is that we have some states $P_{1\mathcal{J}\mathcal{M}}^{0-}$, $P_{1\mathcal{J}\mathcal{M}}^{-0}$, $P_{1\mathcal{J}\mathcal{M}}^{0+}$, $P_{1\mathcal{J}\mathcal{M}}^{+0}$ that are decoupled from the mixing. So their hyperfine spectrum will not change with and without mixing.

Another consideration is that in here we use the corrected signs of [7], so now V^{sa} should be in accordance with [6].

4.1 Potentials

Our potentials with the correct sign are:

$$\begin{aligned} V^{sa}(r) &= + \frac{2\pi^2 |g\Lambda'''| c_F}{m_Q \sigma r^3} \\ V^{sb}(r) &= + \frac{2\pi^{3/2} |g\Lambda'| c_F}{m_Q \sqrt{\sigma} r^2} \end{aligned} \quad (18)$$

This behaviour is extracted from data in [5]. Also using the correct sign of V^{sa} in Eq. (4) in [7] means that the definition of V_{hf} and V_{hf2} is

$$\begin{aligned} V_{hf}(r) &= -\frac{1}{6} V^{sa}(r) - \frac{1}{3} V^{sb}(r) \\ V_{hf2}(r) &= -\frac{1}{2} \left(V^{sb}(r) - V^{sa}(r) \right). \end{aligned} \quad (19)$$

As a first approximation one can use this hypfine potentials at long distances $r > r_0$ with $r_0 = 3.96 \text{ GeV}^{-1}$. And set the hyperfine to zero at short distances.

The next step if to use the interpolation formula

$$\begin{aligned} \frac{V_{hf}}{m_Q} &= \frac{A + \left(\frac{r}{r_0}\right)^2 \left(-\frac{1}{6} V^{sa}(r_0) - \frac{r}{3r_0} V^{sb}(r_0)\right)}{1 + \left(\frac{r}{r_0}\right)^5} \\ \frac{V_{hf2}}{m_Q} &= \frac{B r^2 - \left(\frac{r}{r_0}\right)^5 \left(\frac{1}{2} V^{sb}(r_0) - \frac{r_0}{2r} V^{sa}(r_0)\right)}{1 + \left(\frac{r}{r_0}\right)^7} \end{aligned} \quad (20)$$

With all this set, a first step will be to reproduce Rubens results. For that, instead of trying all sign combinations we juts use + sign for V^{sb} and values $g\Lambda' \simeq 0.1 \text{ GeV}$, $g\Lambda''' \simeq 0.2 \text{ GeV}$, and $\sigma = 0.21 \text{ GeV}^2$. [Should we change the name of \$V_{hf}\$ and \$V_{hf2}\$? Nora said it was confusing.](#)

4.2 Matrices

To the hybrid static potential with mixing (17) we have to add the hyperfine matrices.

$$\Delta V_{\text{hyp}} = -2V_{hf}(r)M_8 - 2V_{hf2}(r)M_9$$

M_8 i M_9 are the matrices in the Ref. [7] with different order: $(P^{--}, P^{+-}, P^{00}, P^{-+}, P^{++})$. This new order implies that for states like $(s/d)_1$ and $(p/f)_2$, the part corresponding to the higher L is in the first position of the 2×1 matrix. Notice also that ΔV_{hyp} is multiplied by m_Q when compared to Ruben notes, this is done in order to take into account that when I compute the Schrödinger equation in **MATLAB** all is consistent.

$$M_8 = \left(\begin{array}{c|ccc} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\mathcal{J}+2}{\mathcal{J}+1} & 0 & 0 & \frac{\mathcal{J}}{\mathcal{J}+1} \frac{\sqrt{2\mathcal{J}+3}}{\sqrt{2\mathcal{J}+1}} \\ 0 & 0 & 0 & -\frac{\mathcal{J}-1}{\mathcal{J}} & 0 & \frac{\mathcal{J}+1}{\mathcal{J}} \frac{\sqrt{2\mathcal{J}-1}}{\sqrt{2\mathcal{J}+1}} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{\mathcal{J}}{\mathcal{J}+1} \frac{\sqrt{2\mathcal{J}+3}}{\sqrt{2\mathcal{J}+1}} & \frac{\mathcal{J}+1}{\mathcal{J}} \frac{\sqrt{2\mathcal{J}-1}}{\sqrt{2\mathcal{J}+1}} & 0 & -\frac{1}{\mathcal{J}(\mathcal{J}+1)} \end{array} \right) \begin{pmatrix} R_{0\mathcal{J}\mathcal{M}} \\ P_{1\mathcal{J}\mathcal{M}}^{++} \\ P_{1\mathcal{J}\mathcal{M}}^{-+} \\ P_{1\mathcal{J}\mathcal{M}}^{+-} \\ P_{1\mathcal{J}\mathcal{M}}^{00} \\ P_{1\mathcal{J}\mathcal{M}}^{++} \end{pmatrix} \quad (21)$$

$$M_9 = - \begin{pmatrix} 0 & | & 0 & 0 & 0 & 0 & 0 \\ 0 & | & -\frac{2(\mathcal{J}+3)}{3(2\mathcal{J}+3)} & V_2^{-+} & 0 & 0 & V_2^{+0} \\ 0 & | & V_2^{-+} & \frac{2\mathcal{J}(\mathcal{J}+2)}{3(\mathcal{J}+1)(2\mathcal{J}+3)} & 0 & 0 & V_2^{00} \\ 0 & | & 0 & 0 & \frac{2(\mathcal{J}^2-1)}{3\mathcal{J}(2\mathcal{J}-1)} & V_2^{--} & V_2^{+-} \\ 0 & | & 0 & 0 & V_2^{--} & -\frac{2(\mathcal{J}-2)}{3(2\mathcal{J}-1)} & V_2^{-0} \\ 0 & | & V_2^{+0} & V_2^{00} & V_2^{+-} & V_2^{-0} & -\frac{2}{3\mathcal{J}(\mathcal{J}+1)} \end{pmatrix} \begin{pmatrix} R_{0\mathcal{J}\mathcal{M}} \\ P_{1\mathcal{J}\mathcal{M}}^{++} \\ P_{1\mathcal{J}\mathcal{M}}^{+-} \\ P_{1\mathcal{J}\mathcal{M}}^{00} \\ P_{1\mathcal{J}\mathcal{M}}^{--} \\ P_{1\mathcal{J}\mathcal{M}}^{-0} \\ P_{1\mathcal{J}\mathcal{M}}^{00} \end{pmatrix} \quad (22)$$

4th-component only exist if $\mathcal{J} > 0$; 5th-component (P^{--}) only exist if $\mathcal{J} > 1$; 6th-component only exist if $\mathcal{J} > 0$.

The matrices above involve the states that are coupled with quarkonium in the mixing. There are also some states in the hyperfine that are decoupled by the mixing with static potential the following

$$m_Q \begin{pmatrix} \frac{(\mathcal{J}+1)(\mathcal{J}+2)}{m_Q r^2} + V_{\Pi_u} & 0 & 0 & 0 \\ 0 & \frac{\mathcal{J}(\mathcal{J}-1)}{m_Q r^2} + V_{\Pi_u} & 0 & 0 \\ 0 & 0 & \frac{(\mathcal{J}+1)(\mathcal{J}+2)}{m_Q r^2} + V_{\Sigma_u^-} + \frac{\mathcal{J}}{2\mathcal{J}+1} V_q & \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} V_q \\ 0 & 0 & \frac{\sqrt{\mathcal{J}(\mathcal{J}+1)}}{2\mathcal{J}+1} V_q & \frac{\mathcal{J}(\mathcal{J}-1)}{m_Q r^2} + V_{\Sigma_u^-} + \frac{\mathcal{J}+1}{2\mathcal{J}+1} V_q \end{pmatrix} \begin{pmatrix} P_{1\mathcal{J}\mathcal{M}}^{0+} \\ P_{1\mathcal{J}\mathcal{M}}^{0-} \\ P_{1\mathcal{J}\mathcal{M}}^{+0} \\ P_{1\mathcal{J}\mathcal{M}}^{-0} \end{pmatrix} \quad (23)$$

2nd-component only exist if $\mathcal{J} > 1$; 3rd-component (P^{--}) and 4th-component only exist if $\mathcal{J} > 0$. Notice that this is static potential order in the $1/m_Q$ expansion so we multiply by m_Q in respect with Rubens notes. For this states the hyperfine structure is

$$\Delta V_{\text{hybrid}} = -2V_{hf}(r)M_{10} - 2V_{hf2}(r)M_{11}$$

on

$$M_{10} = \begin{pmatrix} -\frac{1}{\mathcal{J}+1} & 0 & \frac{\sqrt{\mathcal{J}(\mathcal{J}+2)}}{\mathcal{J}+1} & 0 \\ 0 & \frac{1}{\mathcal{J}} & 0 & \frac{\sqrt{\mathcal{J}^2-1}}{\mathcal{J}} \\ \frac{\sqrt{\mathcal{J}(\mathcal{J}+2)}}{\mathcal{J}+1} & 0 & \frac{1}{\mathcal{J}+1} & 0 \\ 0 & \frac{\sqrt{\mathcal{J}^2-1}}{\mathcal{J}} & 0 & -\frac{1}{\mathcal{J}} \end{pmatrix} \quad (24)$$

$$M_{11} = - \begin{pmatrix} -\frac{2}{3(\mathcal{J}+1)} & 0 & V_2^{+00+} & \frac{\mathcal{J}}{2\mathcal{J}+1} \sqrt{\frac{\mathcal{J}+2}{\mathcal{J}}} \\ 0 & \frac{2}{3\mathcal{J}} & V_2^{0-+0} & V_2^{0--0} \\ V_2^{+00+} & V_2^{0-+0} & -\frac{2(\mathcal{J}+2)}{3(\mathcal{J}+1)(2\mathcal{J}+1)} & V_2^{-0+0} \\ \frac{\mathcal{J}}{2\mathcal{J}+1} \sqrt{\frac{\mathcal{J}+2}{\mathcal{J}}} & V_2^{0--0} & V_2^{-0+0} & -\frac{2}{3\mathcal{J}} + \frac{2}{2\mathcal{J}+1} \end{pmatrix} \quad (25)$$

Notice that we have changed the order to be in agreement with the new ordering of (21) and (22).

In summary if we want to compute:

- Hyperfine potentials *without* mixing: The matrices will not involve the quarkonium state $R_{0\mathcal{J}\mathcal{M}}$. To the states $P_{1\mathcal{J}\mathcal{M}}^{++}$, $P_{1\mathcal{J}\mathcal{M}}^{+-}$, $P_{1\mathcal{J}\mathcal{M}}^{+0}$, $P_{1\mathcal{J}\mathcal{M}}^{00}$, and $P_{1\mathcal{J}\mathcal{M}}^{00}$ we need to add Matrix (17) (without quarkonium) and Matrices M_8 and M_9 together. To the states $P_{1\mathcal{J}\mathcal{M}}^{0-}$, $P_{1\mathcal{J}\mathcal{M}}^{-0}$, $P_{1\mathcal{J}\mathcal{M}}^{+0}$, and $P_{1\mathcal{J}\mathcal{M}}^{0+}$ we need to add Matrix (23) and Matrices M_{10} and M_{11} together.
- Hyperfine potentials *with* mixing: In here we will involve the hole Matrix (17) and the rest remains the same as the previous point.

5 Code implementation

With this done by Ruben in his files `VSigG.wl`, `VPiU.wl`, `VSigS.wl`, `VSigMix.wl` and `VPiMix.wl` we can directly implement it in `MATLAB` as functions.

5.1 Files

Each physics file contains its own potential matrix (`potentialMatrix`) and potentials. The number of channels is read from the potential matrix, so it follows the `mix` flag automatically. Units are GeV and GeV^{-1} .

- `QQbarS0J{0,1,2}.m`: quarkonium spin 0 ($\mathcal{J} = J = L$). With `mix` they solve the `MixingH1` matrix (17) (quarkonium spin 0 with hybrid spin 1, without hyperfine terms). Without `mix` we solve Eq. (14) or the same as with `mix` without the P states.
- `QQbarS1J{0,1,2}.m`: quarkonium spin 1. They are only implemented with `mix = true`, where they solve the `MixingH0` matrices (15) and (16).
- `GQQbarS0Jcal{0,1,2}.m`: hybrids spin 0. With `mix` they solve the same `MixingH0` matrix as `QQbarS1J*.m`, so both give identical spectra. Without `mix` we solve only the hybrid part of (15) and (16).
- `GQQbarS1Jcal{0,1,2,3}.m`: hybrids spin 1 with the hyperfine terms, they depend on A and B . With `mix` they solve the `MixingH1` matrix (17) plus M_8 , M_9 , and the decoupled block (23) plus M_{10} , M_{11} . Without `mix` we solve the same without R states.
- `quarkoniumConfig.m` and `hybridConfig.m`: all the flags in one place: `Eg`, `Eval`, `mix` (both), and `hf`, `glamb1`, `glamb3`, `sig`, `pm`, `r0` (hybrids). They can be changed in a script, e.g. `hybridConfig('mix',true)`, and are restored with `hybridConfig('reset')`.
- `solveRadialSchrodinger.m`, `radialMesh.m`, `radialHamiltonian.m`: solver of the coupled-channel radial equation $-u'' + Vu = mEu$ (see Sec. 5.2).
- `QQbarSpectrum.m`: computes every family and plots the wave functions.
- `QQbarSpectrumTable.m`: computes every family and assigns to each state J^{PC} , its dominant component and the percentages of quarkonium and hybrid, in the format of Ruben's Tables 5 and 9 (Tables 2 and 4 below).
- `hybridAssignments.m`: has the lattice data $t \pm e$ and the assignation of the computed states, one table for `mix = false` and one for `mix = true`.
- `QQbarSpectrumOptimize.m`: fit of A and B to the lattice hybrid spectrum, with their errors and the errors of the fitted energies (Section 5.9).
- `spinAverages.m`: spin averages and splittings of each multiplet compared with the data, with a χ^2 breakdown.
- `proposeMixAssignment.m`: proposes the `mix = true` assignment table from the `mix = false` one (overlap of the wave functions), as a cross-check. **The final assignation is done by hand so I was just a helpful try.**

5.2 Numerical method

`solveRadialSchrodinger.m` uses a regularized Lagrange–Laguerre mesh [1]. The radial functions behave as r at the origin and decay exponentially at large r . The whole coupled-channel problem $-u''(r) + V(r)u(r) = mEu(r)$ becomes one symmetric matrix and a single diagonalization gives every level and wave function at once: there is no shooting or matching, no NaNs, degenerate levels come out correctly and the energies depend smoothly on the parameters. The default mesh has $N = 100$ points per channel with the last point at $r_{\max} = 25 \text{ GeV}^{-1}$ (the fit uses $N = 60$); the energies are converged to $\sim 10^{-6} \text{ GeV}$. This replaces Ruben’s shooting solver, which sometimes gave NaNs.

The channel weights $\int |u_c|^2 dr$ of each state give its composition: $\%Q$ is the sum over the quarkonium channels (R_0 in `MixingH1`; R_1^0 , $R_1^\pm$ in `MixingH0`) and $\%H = 100 - \%Q$.

5.3 Static Spectrum Results for Charmonium

For **quarkonium spin 0**, we execute the codes `QQbarS0Jn.m` with $n = 0, 1, 2$ depending if we want to obtain the S-wave, P-wave or D-wave spectrum with the `mix` parameter set to `false`. In here $\mathcal{J} = J = L$.

The energy eigenvalues E obtained directly from this Hamiltonian do not yet correspond to the physical masses. The gluonic shift E_g must be added to the fitted potentials ($V_{\Sigma_g^+}$, V_{Π_u} , and $V_{\Sigma_u^-}$) to match the experimental physical scale. Therefore, to obtain the proper physical mass of the states, we must shift the raw values by the mass of the constituent quarks and this gluonic shift:

$$M_{Q\bar{Q}} = 2m_Q + E + E_g. \quad (26)$$

The unshifted energy eigenvalues E for the principal quantum numbers $n = 1, \dots, 7$ are summarized in Table 1 which by comparing to Fig. 5 in Ruben’s notes we can see that the potentials are well implemented. After using following Ruben’s $m_Q = 1.496 \text{ GeV}$ and $E_g = 0.45 \text{ GeV}$, the spectrum agrees with the charmonium one from [3]. This change is implemented changing the `Eg` parameter to `true`.

n	s states ($J = 0$)	p states ($J = 1$)	d states ($J = 2$)
1	0.553049	0.939710	1.239707
2	1.151416	1.430948	1.677433
3	1.616675	1.851244	2.067372
4	2.020772	2.228958	2.424919
5	2.386970	2.577236	2.758541
6	2.726438	2.903473	3.073497
7	3.045615	3.212339	3.373322

Table 1: Unshifted energy eigenvalues E in GeV for charmonium s , p , and d states ($n = 1, \dots, 7$), without mixing, computed with the Lagrange–Laguerre solver. These raw values require the shift $2m_c + E_g$ to match the physical spectrum. They are 6–20 MeV below the values obtained with the shooting solver (good agreement).

For **quarkonium spin 1** spectrum, we obtain the same values because at this order they are degenerated with quarkonium spin 0.

5.4 Mixing Results for Charmonium

For **quarkonium spin 0**, we use the same codes `QQbarS0Jn.m` with $n = 0, 1, 2$ but now with `mix` parameter set to `true`. In here we still have $\mathcal{J} = J = L$.

For **quarkonium spin 1**, we use the scripts `QQbarS1Jn.m` with $n = 0, 1, 2$ where the parameter `mix` is always `true`. However, now, $\mathcal{J} = J \neq L$. Therefore $n = 0$ gives P-states; $n = 1$ gives either S, P, or D-states; and $n = 2$ is for P and D-states. Therefore the spectrum obtained is not ordered. And to obtain an ordered spectrum we need to check for each state.

Table 2 gives all the states below $E = 1.8$ GeV obtained with `QQbarSpectrumTable.m` from the quarkonium files, in the format of Ruben's Table 5. The label of each state (composition, S, j, L and J^{PC}) is the one of its dominant channel, with $P = (-1)^{L+1}$ and $C = c_g(-1)^{L+S}$ ($c_g = +1$ quarkonium, -1 hybrid). n is the position of the state in the energy order of its file.

Comparison with Ruben's Table 5:

- Every state of Table 5 is found with the same S, J, j, L, J^{PC} and dominant component.
- `MixingH1 (QQbarSOJ*.m)`: the hybrid-dominated states agree within 1 MeV (1513, 1661, 1752 MeV). The R_0 states are 1–10 MeV lower (e.g. 1094 vs. 1098, 1389 vs. 1399 MeV) with the same percentages. The 0^{-+} pair near 1580 MeV is an avoided crossing: the energies agree within 4 MeV, but we obtain an almost 50/50 mixture (1572: 55% Q, 1585: 47% Q) while Table 5 has 1575 (33% Q, P_1^{-+}) and 1589 (70% Q, R_0).
- `MixingH0 (QQbarS1J*.m)`: the $R_1^{\pm}$ and $P_0^{\pm}$ states agree within 1–9 MeV, closer to the values of Ruben's Table 9 than to Table 5 (both tables should coincide for `MixingH0`, since the hyperfine terms only act on spin-1 hybrids).
- The only clear disagreement is the $R_1^0-P_0^0$ pair of Eq. (15): we get 1^{++} at 920, 1418 MeV and 2^{--} at 1228, 1668 MeV (2% H), while Table 9 has 904, 1410, 1215, 1660 MeV (3–4% H); the P_0^0 hybrid 1^{++} is at 1666 MeV instead of 1678 MeV. Multiplying the coupling $2V_S^{\Pi}$ of Eq. (15) by $\sqrt{2}$ reproduces Table 9: 900, 1405, 1679 MeV (1^{++}) and 1216, 1659 MeV (2^{--}). **Ruben's code seems to use $2V_S^{\Sigma}$ in this element: check whether Eq. (15) or his implementation is correct. I used $2V_S^{\Pi}$.**

5.5 Static Spectrum Results for Charm Hybrids

The static spectrum of **hybrids spin 0** is degenerate to **hybrids spin 1**. Therefore to obtain the hybrid spectrum like the one in Table I of [3] we use the potentials in Eq. (6) and (7) of that same paper which correspond to (15) and (16) respectively without the quarkonium states. We use the codes `GQQbarSOJcaln.m` with $n = 0, 1, 2$ with the parameter `mix` set as `false`. These codes give the states classified for $\mathcal{J}$

- $n = 0$ gives p_0 -states
- $n = 1$ gives both $(s/d)_1$ and p_1 -states
- $n = 2$ gives both $(p/f)_2$ and d_2 -states

Hence the shape of the wave function changes with the index of the array for $n = 1, 2$.

As we do not include the hyperfine splitting here, hybrids with spin 1 are degenerate with hybrids of spin 0.

If we set `Eg` as `true` we get results very similar to those of Table I from [3].

5.6 Mixing Results for Charm Hybrids

For **hybrids spin 0** we use the codes `GQQbarSOJcaln.m` with $n = 0, 1, 2$ with the parameter `mix` set as `true`.

For **hybrids spin 1** we use the codes `GQQbarS1Jcaln.m` with $n = 0, 1, 2$ with the parameter `mix` set as `true` but the parameter `hf` set to `false`. We most definitely will not use this case but just in case its included. The spectrum should be degenerated to the one with hybrids with spin 0 however, the mixing can break slightly this degeneracy.

Matrix	n	S	J_{Tot}	j	L	E (MeV)	M (MeV)	J^{PC}	Comp.	Type	%Q	%H
MixingH1	1	0	0	0	0	441	2983	0^{-+}	R_0	Quarkonium	91	9
MixingH0	1	1	1	1	0	514	3056	1^{--}	R_1^-	Quarkonium	97	3
MixingH1	1	0	1	1	1	873	3415	1^{+-}	R_0	Quarkonium	93	7
MixingH0	1	1	0	0	1	911	3453	0^{++}	R_1^+	Quarkonium	97	3
MixingH0	1	1	2	2	1	916	3458	2^{++}	R_1^-	Quarkonium	98	2
MixingH0	2	1	1	1	1	920	3462	1^{++}	R_1^0	Quarkonium	98	2
MixingH1	2	0	0	0	0	1094	3636	0^{-+}	R_0	Quarkonium	95	5
MixingH0	3	1	1	1	0	1131	3673	1^{--}	R_1^-	Quarkonium	98	2
MixingH1	1	0	2	2	2	1199	3741	2^{-+}	R_0	Quarkonium	95	5
MixingH0	4	1	1	1	2	1224	3766	1^{--}	R_1^+	Quarkonium	98	2
MixingH0	2	1	2	2	2	1228	3770	2^{--}	R_1^0	Quarkonium	98	2
MixingH1	2	0	1	1	1	1389	3931	1^{+-}	R_0	Quarkonium	96	4
MixingH0	2	1	0	0	1	1413	3955	0^{++}	R_1^+	Quarkonium	98	2
MixingH0	3	1	2	2	1	1416	3958	2^{++}	R_1^-	Quarkonium	98	2
MixingH0	5	1	1	1	1	1418	3960	1^{++}	R_1^0	Quarkonium	98	2
MixingH0	4	1	2	2	3	1492	4034	2^{++}	R_1^+	Quarkonium	99	1
MixingH1	2	1	2	1	0	1513	4055	2^{-+}	P_1^{--}	Hybrid	0	100
MixingH0	6	0	1	1	0	1538	4080	1^{--}	P_0^-	Hybrid	2	98
MixingH1	3	0	0	0	0	1572	4114	0^{-+}	R_0	Quarkonium	55	45
MixingH1	4	1	0	1	0	1585	4127	0^{-+}	P_1^{-+}	Hybrid	47	53
MixingH0	7	1	1	1	0	1603	4145	1^{--}	R_1^-	Quarkonium	99	1
MixingH1	3	0	2	2	2	1647	4189	2^{-+}	R_0	Quarkonium	96	4
MixingH1	3	1	1	1	1	1661	4203	1^{+-}	P_1^{00}	Hybrid	1	99
MixingH0	8	1	1	1	2	1665	4207	1^{--}	R_1^+	Quarkonium	99	1
MixingH0	9	0	1	1	1	1666	4208	1^{++}	P_0^0	Hybrid	2	98
MixingH0	5	1	2	2	2	1668	4210	2^{--}	R_1^0	Quarkonium	98	2
MixingH0	6	0	2	2	1	1742	4284	2^{++}	P_0^-	Hybrid	2	98
MixingH1	4	1	1	2	1	1752	4294	1^{+-}	P_1^{-+}	Hybrid	3	97

Table 2: Charmonium spectrum from the quarkonium files QQbarS0J*.m (MixingH1) and QQbarS1J*.m (MixingH0) with `mix = true`, obtained with QQbarSpectrumTable.m. E is the Schrödinger energy and $M = E + 2m_c - E_{\text{val}}$, with $m_c = 1.496$ GeV, $E_{\text{val}} = 0.45$ GeV and $c_F = 1.12$. To compare with Ruben's Table 5.

5.7 Hyperfine Results for Charm Hybrids

This subsection only applies for **hybrids spin 1** as the hyperfine splitting breaks their degeneration to spin 0 hybrids. We will differentiate between the results *with* mixing and *without* mixing. In principle, the last should give a similar spectrum than the computed in the paper [7] however, as our potentials and also values for σ have changed, the spectrum can change.

As a first approximation I'm going to do all the calculations with the long distance shape of the hyperfine potentials (19) and using Ruben's values for the constants involved. So we will obtain different results from that Reference.

The code used is once again `GQQbarS1Jcaln.m` with $n = 0, 1, 2$ with the parameter `hf` set as `true`.

If we set the `mix` to `true` for both spin 0 and 1 and `hf` set to `true` for the latter, we need to be careful assigning the states. Because for example $n = 1$ in code `GQQbarS1Jcaln.m` will plot all the states with $\mathcal{J} = 1$ which means several kinds of hybrids such as p_0 , $(s/d)_1$, p_1 , $(p/f)_2$, and d_2 . To assign a state we must look at their wave function to be able to characterize them. This is now done automatically by `QQbarSpectrumTable.m` from the channel weights: Table 4 gives all the states below $E = 1.8$ GeV of the hybrid files with `mix = true` and `hf = true`, using the interpolation (20) with the best-fit values with mixing, $A = -0.085543$ GeV and $B = -0.016410$ GeV³ (Section 5.9). It also contains the spin-1 hybrids decoupled from quarkonium, Eq. (23), which are not listed in Ruben's Table 9 (e.g. 1402 MeV P_1^{-0} , 1^{-+} , and 1619 MeV P_1^{0+} , 0^{+-}).

Comparison with Ruben's Table 9:

- `MixingH0 (GQQbarS0Jcal*.m)`: same states as in Table 2; see the comments there.
- `MixingH1 quarkonium-dominated states (GQQbarS1Jcal*.m)`: within 6–14 MeV, e.g. 415 vs. 425, 852 vs. 866, 1184 vs. 1195, 1468 vs. 1474 MeV, with a few percent more hybrid component (they are pushed down by the hybrid levels, which lie lower for the fitted A , B).
- `MixingH1 hybrid-dominated states`: they differ by 36–98 MeV, see Table 3. This is expected, as Table 9 was computed with the long distance versions of V_{hf} and V_{hf2} potentials (19) (**and four wrong terms in the matrices $M9$ and $M11$**), while here we use the interpolation (20) with A and B fitted with mixing. With the no-mix values $A = -0.048430$ GeV, $B = -0.009419$ GeV³ the differences are 14–49 MeV: the larger $|A|$ of the mix fit increases the hyperfine splittings, lowering the 0^{-+} and 1^{+-} states and raising the 2^{-+} and 3^{+-} ones. The stronger mixing of the quarkonium states comes from our hybrid levels lying lower.

File	n	J^{PC}	Comp.	%Q / %H	E (MeV)	Table 9	Δ
GQQbarS1Jcal0	3	0^{-+}	P_1^{-+}	10 / 90	1386	1484	-98
GQQbarS0Jcal1	6	1^{--}	P_0^-	2 / 98	1538	1538	0
GQQbarS1Jcal2	2	2^{-+}	P_1^{-+}	0 / 100	1603	1567	+36
GQQbarS1Jcal1	4	1^{+-}	P_1^{00}	8 / 92	1584	1644	-60
GQQbarS0Jcal1	9	1^{++}	P_0^0	2 / 98	1666	1678	-12
GQQbarS1Jcal1	5	1^{+-}	P_1^{-+}	2 / 98	1606	1698	-92
GQQbarS0Jcal2	6	2^{++}	P_0^-	2 / 98	1742	1742	0
GQQbarS1Jcal3	2	3^{+-}	P_1^{-+}	0 / 100	1808	1763	+45

Table 3: Hybrid-dominated states of Ruben's Table 9 (red) compared with our results (`mix = true`, `hf = true`, $A = -0.085543$ GeV, $B = -0.016410$ GeV³). n is the position of the state in the energy order of its file. The 3^{+-} state is just above the 1.8 GeV cut of Table 4. The spin-0 hybrids (`GQQbarS0Jcal*`) have no hyperfine terms and agree; the 1678 MeV difference is the $R_1^0-P_0^0$ coupling discussed in Table 2.

Matrix	n	S	J_{Tot}	j	L	E (MeV)	M (MeV)	J^{PC}	Comp.	Type	%Q	%H
MixingH1	1	0	0	0	0	415	2957	0^{-+}	R_0	Quarkonium	87	13
MixingH0	1	1	1	1	0	514	3056	1^{--}	R_1^-	Quarkonium	97	3
MixingH1	1	0	1	1	1	852	3394	1^{+-}	R_0	Quarkonium	89	11
MixingH0	1	1	0	0	1	911	3453	0^{++}	R_1^+	Quarkonium	97	3
MixingH0	1	1	2	2	1	916	3458	2^{++}	R_1^-	Quarkonium	98	2
MixingH0	2	1	1	1	1	920	3462	1^{++}	R_1^0	Quarkonium	98	2
MixingH1	2	0	0	0	0	1081	3623	0^{-+}	R_0	Quarkonium	92	8
MixingH0	3	1	1	1	0	1131	3673	1^{--}	R_1^-	Quarkonium	98	2
MixingH1	1	0	2	2	2	1184	3726	2^{-+}	R_0	Quarkonium	92	8
MixingH0	4	1	1	1	2	1224	3766	1^{--}	R_1^+	Quarkonium	98	2
MixingH0	2	1	2	2	2	1228	3770	2^{--}	R_1^0	Quarkonium	98	2
MixingH1	2	0	1	1	1	1377	3919	1^{+-}	R_0	Quarkonium	93	7
MixingH1	3	1	0	1	0	1386	3928	0^{-+}	P_1^{-+}	Hybrid	10	90
MixingH1	3	1	1	1	0	1402	3944	1^{-+}	P_1^{-0}	Hybrid	0	100
MixingH0	2	1	0	0	1	1413	3955	0^{++}	R_1^+	Quarkonium	98	2
MixingH0	3	1	2	2	1	1416	3958	2^{++}	R_1^-	Quarkonium	98	2
MixingH0	5	1	1	1	1	1418	3960	1^{++}	R_1^0	Quarkonium	98	2
MixingH1	1	0	3	3	3	1468	4010	3^{+-}	R_0	Quarkonium	94	6
MixingH0	4	1	2	2	3	1492	4034	2^{++}	R_1^+	Quarkonium	99	1
MixingH0	6	0	1	1	0	1538	4080	1^{--}	P_0^-	Hybrid	2	98
MixingH1	4	0	0	0	0	1569	4111	0^{-+}	R_0	Quarkonium	95	5
MixingH1	4	1	1	1	1	1584	4126	1^{+-}	P_1^{00}	Hybrid	8	92
MixingH0	7	1	1	1	0	1603	4145	1^{--}	R_1^-	Quarkonium	99	1
MixingH1	2	1	2	1	0	1603	4145	2^{-+}	P_1^{-}	Hybrid	0	100
MixingH1	5	1	1	2	1	1606	4148	1^{+-}	P_1^{-+}	Hybrid	2	98
MixingH1	3	1	2	1	1	1612	4154	2^{+-}	P_1^{0-}	Hybrid	0	100
MixingH1	5	1	0	1	1	1619	4161	0^{+-}	P_1^{0+}	Hybrid	0	100
MixingH1	4	0	2	2	2	1634	4176	2^{-+}	R_0	Quarkonium	91	9
MixingH0	8	1	1	1	2	1665	4207	1^{--}	R_1^+	Quarkonium	99	1
MixingH0	9	0	1	1	1	1666	4208	1^{++}	P_0^0	Hybrid	2	98
MixingH0	5	1	2	2	2	1668	4210	2^{--}	R_1^0	Quarkonium	98	2
MixingH1	5	1	2	2	1	1724	4266	2^{+-}	P_1^{-0}	Hybrid	0	100
MixingH0	6	0	2	2	1	1742	4284	2^{++}	P_0^-	Hybrid	2	98
MixingH1	6	1	2	2	2	1799	4341	2^{-+}	P_1^{00}	Hybrid	9	91

Table 4: Charmonium spectrum from the hybrid files `GQQbarS0Jcal*.m` (MixingH0) and `GQQbarS1Jcal*.m` (MixingH1 with hyperfine terms) with `mix = true` and `hf = true`, at the best fit with mixing $A = -0.085543$ GeV and $B = -0.016410$ GeV³, obtained with `QQbarSpectrumTable.m`. Same conventions as Table 2. To compare with Ruben's Table 9.

5.8 Spectrum

Once the individual codes are done we can compute the spectrum and the wave functions for each kind of quarkonium and hybrid using `QQbarSpectrum.m`.

First, we used the long distance shape of the hyperfine potentials (19) with V_{hf} and V_{hf2} zero for $r < r_0 = 3.96 \text{ GeV}^{-1}$. Then, we can use the full interpolation (20) with A and B the ones found in [7]. Finally, we obtain new A and B setting the `mix` as `true`.

To obtain the good values for A and B we use the script `QQbarSpectrumOptimize.m` (this also gives the uncertainties assuming lattice data to be fully uncorrelated). The lattice data used is the same as in [7].

It is useful to remember how the states are written in order to make identifications.

- The lattice data $t \pm e$ and the state compared with each data point are in `hybridAssignments.m`. The order of the data is four $(s/d)_1$ states, four p_1 states, four $(p/f)_2$ states, and two p_0 states. Each data point is assigned to $\{S, \mathcal{J}, n\}$: the n -th state in energy order of `GQQbarS<S>Jcal<J>.m` at the best fit, whose (A, B) is stored with the table (`ref`). There is one assignment table for `mix = false` and one for `mix = true`, since the quarkonium states change the indices. With mixing the quarkonium levels lie between the hybrid ones, so some indices change with A and B even if the state is the same (see the mix best fit below), and they must always be checked with the channel weights.
- Hybrid states with spin 0 are ordered $(P_{0\mathcal{J}\mathcal{M}}^0 P_{0\mathcal{J}\mathcal{M}}^+ P_{0\mathcal{J}\mathcal{M}}^-)$. If we include the mixing then it reads $(R_{1\mathcal{J}\mathcal{M}}^0 P_{0\mathcal{J}\mathcal{M}}^0 R_{1\mathcal{J}\mathcal{M}}^+ R_{1\mathcal{J}\mathcal{M}}^- P_{0\mathcal{J}\mathcal{M}}^+ P_{0\mathcal{J}\mathcal{M}}^-)$. Remember that for $\mathcal{J} = 0$ only the $+$ states exist.
- Hybrid states with spin 1 are ordered as follows (with mixing, $R_{0\mathcal{J}\mathcal{M}}$ is added in the first position)
 1. $\mathcal{J} = 0$: $(P_{1\mathcal{J}\mathcal{M}}^{++} P_{1\mathcal{J}\mathcal{M}}^{-+} P_{1\mathcal{J}\mathcal{M}}^{0+})$
 2. $\mathcal{J} = 1$: $(P_{1\mathcal{J}\mathcal{M}}^{++} P_{1\mathcal{J}\mathcal{M}}^{-+} P_{1\mathcal{J}\mathcal{M}}^{+-} P_{1\mathcal{J}\mathcal{M}}^{00} P_{1\mathcal{J}\mathcal{M}}^{0+} P_{1\mathcal{J}\mathcal{M}}^{+0} P_{1\mathcal{J}\mathcal{M}}^{-0})$
 3. $\mathcal{J} > 1$: $(P_{1\mathcal{J}\mathcal{M}}^{++} P_{1\mathcal{J}\mathcal{M}}^{-+} P_{1\mathcal{J}\mathcal{M}}^{+-} P_{1\mathcal{J}\mathcal{M}}^{--} P_{1\mathcal{J}\mathcal{M}}^{00} P_{1\mathcal{J}\mathcal{M}}^{0+} P_{1\mathcal{J}\mathcal{M}}^{0-} P_{1\mathcal{J}\mathcal{M}}^{+0} P_{1\mathcal{J}\mathcal{M}}^{-0})$

The identification of the states can be checked with the plots of the wave functions in `QQbarSpectrum.m`, with Tables 2 and 4, and with the assignment table printed by `QQbarSpectrumOptimize.m`, which gives for each data point the channel block and the energy at the best fit.

5.9 Fit of A and B

- **Model.** The spin-1 hybrid Hamiltonian is exactly linear in A and B , $H(A, B) = H_0 + A H_A + B H_B$, so H_0 , H_A , H_B are built once and each (A, B) only needs a diagonalization. The spin-0 hybrids do not depend on A and B .
- **Minimization.** $\chi^2 = \sum_k (t_k - E_k)^2 / e_k^2$ is scanned on a 25×25 grid in $A \in [-0.3, 0.3] \text{ GeV}$, $B \in [-0.06, 0.06] \text{ GeV}^3$ for the starting point and then minimized with `lsqnonlin` using the exact Jacobian from Hellmann–Feynman, $\partial E / \partial A = \langle \psi | H_A | \psi \rangle / m_Q$. Each data point is identified once, at the `ref` point of the assignment table, as the k -th level of its decoupled channel block (the mixing block with R_0 , $P_1^{\pm\pm}$, P_1^{00} , or the block $P_1^{0\pm}$, $P_1^{\pm 0}$ of Eq. (23)), and this label is followed in the grid scan and in the fit. Levels of the same block do not cross, while levels of different blocks can cross without being swapped. At the minimum the energy order is checked again: if it changed, the script prints the new indices to update the table.
- **Errors of A and B .** $C = (J^T J)^{-1}$ at the minimum ($\Delta\chi^2 = 1$), scaled by χ^2/dof when $\chi^2/\text{dof} > 1$ (PDG convention). These scaled errors are the ones given in Table 5. J here is the Jacobian.

- **Errors of the fitted energies.** The covariance of (A, B) , including their correlation ρ , is propagated to each energy,

$$\sigma_{\text{fit}}^2 = G C G^T, \quad G = \left(\frac{\partial E}{\partial A}, \frac{\partial E}{\partial B} \right), \quad (27)$$

and checked beyond linear order: σ_{fit} is taken as the largest $|E - E_{\text{best}}|$ on the 1σ ellipse of (A, B) (for E linear in A, B both give exactly the same, here they agree within 2 MeV). Taking the four corners $(A \pm \sigma_A, B \pm \sigma_B)$, as we did on [7], instead would ignore the correlation and always overestimate the error. σ_{fit} measures how much a state depends on A and B , so it vanishes for the spin-0 hybrids. Finally, the higher orders are included adding in quadrature $\sigma_{\text{th}} = 0.03$ GeV,

$$\sigma_E = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{th}}^2}. \quad (28)$$

σ_{th} is only added after the fit; it could also be included in the χ^2 as $\sqrt{e^2 + \sigma_{\text{th}}^2}$, which would change A, B and their errors.

The results are in Table 5 and Figs. 2 and 3. They are computed using the interpolations (20) with $r_0 = 3.96$ GeV $^{-1}$; gluonic shifts in the spectrum given by $M_{Q\bar{Q}} = E + 2m_Q - E_{\text{val}}$ with $m_Q = 1.496$ GeV and $E_{\text{val}} = 0.45$ GeV; $\sigma = 0.21$ GeV 2 , $|g\Lambda''| \simeq 0.2$ GeV and $|g\Lambda'| \simeq 0.1$ GeV.

- **Paper:** A and B of [7] in our code without mixing, $\chi^2/\text{dof} = 79.1/12 = 6.59$ (with mixing it would be $71.4/12 = 5.95$).
- **No mix best:** $A = -0.0484 \pm 0.0115$ GeV, $B = -0.0094 \pm 0.0064$ GeV 3 , $\rho = 0.46$, $\chi^2/\text{dof} = 27.2/12 = 2.27$ (p -value 0.007). The correct assignment of the $(p/f)_2$ 2^{+-} state (GQQbarS1Jcal2 $n = 3$, the second level of the decoupled block $P_1^{0\pm}, P_1^{\pm 0}$, instead of $n = 4$) reduces its pull from -2.9 to -1.3 and the χ^2 from 33.0 to 27.2. The four spin-0 points, which do not depend on A and B , contribute 7.5 to the χ^2 . The largest pulls are the $(p/f)_2$ 3^{+-} (-2.4), the $(s/d)_1$ 2^{-+} (-2.2) and 1^{-+} (-2.2) and the p_1 1^{++} (-2.0 , spin 0); with σ_E added to the lattice error no pull is larger than 1.8. **Maybe we should check the fit of $V_{\Sigma_u^-}$ potentials because is the one responsible for $(s/d)_1$ and $(p/f)_2$ states associated with P^+, P^- .**
- **Mix best:** $A = -0.0855 \pm 0.0163$ GeV, $B = -0.0164 \pm 0.0080$ GeV 3 , $\rho = 0.40$, $\chi^2/\text{dof} = 47.7/12 = 3.97$ (p -value 4×10^{-6}). With mixing the spin-0 points move up by 14–26 MeV and contribute 18.3 to the χ^2 (they do not depend on A and B), and the fit moves to a larger $|A|$. The largest pulls are the $(s/d)_1$ 2^{-+} (-3.2), the $(p/f)_2$ 3^{+-} (-2.9), the $(s/d)_1$ 1^{--} (-2.9 , spin 0) and the p_1 1^{++} (-2.5 , spin 0); with σ_E added to the lattice error no pull is larger than 2.1.

The assignment with mixing needs care, because the quarkonium levels lie between the hybrid ones and several levels are close in energy, so the energy index of a state changes with A and B . At the best fit (checked with the channel weights):

- p_1 0^{+-} (P_1^{0+} , decoupled) is GQQbarS1Jcal0 $n = 5$, since it is now above the R_0 $3S$ quarkonium level ($n = 4$, 95% Q). At $A = -0.0807$ it was $n = 4$.
- $(s/d)_1$ 2^{-+} (P_1^{-+} , mixing block) and p_1 2^{+-} (P_1^{0-}, P_1^{-0} , decoupled) are GQQbarS1Jcal2 $n = 2$ and $n = 3$, only 9 MeV apart. Their energy order changes around $A \simeq -0.09$ (at $A = -0.0807$ they were $n = 3$ and $n = 2$).
- $(p/f)_2$ 3^{+-} (P_1^{--} , mixing block) is GQQbarS1Jcal3 $n = 2$, 11 MeV below a decoupled level.

If the states are followed only by energy order, the fit jumps between these solutions and can end at spurious minima with a lower χ^2 (42–43) in which, e.g., the p_1 0^{+-} lattice point is compared with the $\eta_c(3S)$ quarkonium level. This is why the fit now follows each state by its channel block from the **ref** point. With the physical identification the minimum is unique (the same from different starting points) and the energy order at the minimum is the one of the table.

From **spinAverages.m** (Table 6), the spin averages of the $(s/d)_1$ and $(p/f)_2$ multiplets are 48 and 50 MeV above the lattice ones (the p_1 and p_0 averages agree within 15 MeV), and the four spin averages alone give 27.6 of the 47.7. More than half of the χ^2 is therefore not in the splittings (the part controlled by A and B) but in the position of the $(s/d)_1$ and $(p/f)_2$ multiplets as a whole.

Charm Spectrum		Lattice	Paper	No mix Best	Mix Best
A (GeV)			−0.070	−0.0484(115)	−0.0855(163)
B (GeV ³)			0.0117	−0.0094(64)	−0.0164(80)
$\rho(A, B)$				0.46	0.40
χ^2/dof			6.59	2.27	3.97
$(s/d)_1$ mass (GeV)	1^{--}	4.030(18)	4.055	4.055(30)	4.080(30)
	0^{-+}	3.898(19)	3.837	3.903(37)	3.928(41)
	1^{-+}	3.929(24)	3.938	3.980(32)	3.944(34)
	2^{-+}	4.075(22)	4.132	4.123(32)	4.145(34)
p_1 mass (GeV)	1^{++}	4.144(26)	4.194	4.194(30)	4.208(30)
	0^{+-}	4.111(28)	4.021	4.151(43)	4.161(49)
	1^{+-}	4.112(24)	4.032	4.108(34)	4.126(35)
	2^{+-}	4.176(19)	4.129	4.173(31)	4.154(32)
$(p/f)_2$ mass (GeV)	2^{++}	4.231(33)	4.269	4.269(30)	4.284(30)
	1^{+-}	4.179(28)	4.211	4.183(35)	4.148(39)
	2^{+-}	4.239(27)	4.356	4.272(37)	4.266(42)
	3^{+-}	4.252(35)	4.319	4.333(31)	4.350(31)
p_0 mass (GeV)	0^{++}	4.440(47)	4.465	4.465(30)	4.491(30)
	1^{+-}	4.514(54)	4.456	4.478(30)	4.493(30)

Table 5: Mass spectrum comparison across different parameters A and B . Lattice: data used in the fit, $t(e)$. Paper: our code with A and B of [7], without mixing. No mix Best: best fit without mixing with quarkonium. Mix Best: best fit with mixing. The errors of A and B are scaled by $\sqrt{\chi^2/\text{dof}}$; the errors of the masses are $\sigma_E = \sqrt{\sigma_{\text{fit}}^2 + (0.03 \text{ GeV})^2}$. The $(s/d)_1$ 1^{--} , p_1 1^{++} , $(p/f)_2$ 2^{++} and p_0 0^{++} entries are spin-0 hybrids, independent of A and B (only σ_{th}).

5.10 Spin averages and pulls

To separate what the static potentials do from what the hyperfine terms (A , B) do, **spinAverages.m** computes for each multiplet $(s/d)_1$, p_1 , $(p/f)_2$, p_0 (the spin-0 state and the spin-1 states with the same light angular momentum j and parity) the spin average

$$\overline{M} = \sum_J w_J M_J, \quad w_J = \frac{2J+1}{\sum_{J'} (2J'+1)}, \quad (29)$$

both for the lattice data and for the model at the best fit. For a complete multiplet the weights $2J+1$ count the states, so the spin-dependent (hyperfine) terms cancel in $\overline{M}$ at first order in perturbation theory: $\overline{M}$ tests the static potentials and E_{val} , while the splittings $M_J - \overline{M}$ inside the multiplet test A and B . The lattice errors are taken as uncorrelated, as in the fit, so the

error of the lattice spin average is

$$\sigma_{\overline{M}} = \sqrt{\sum_J w_J^2 e_J^2}. \quad (30)$$

The **pull** is the difference between lattice and model in units of the lattice error,

$$\text{pull}_k = \frac{t_k - E_k}{e_k} \quad (\text{each data point}), \quad \text{pull}(\overline{M}) = \frac{\overline{M}_{\text{lat}} - \overline{M}_{\text{model}}}{\sigma_{\overline{M}}} \quad (\text{each multiplet}). \quad (31)$$

A pull of -3 means that the model is 3σ *above* the lattice value (negative: model too heavy; positive: model too light). If the model were correct and the errors Gaussian and uncorrelated, the pulls would be distributed as a normal distribution of mean 0 and width 1: $|\text{pull}| \lesssim 1$ is good agreement and $|\text{pull}| \gtrsim 3$ is a clear tension. The χ^2 of the fit is the sum of the squared pulls of the data points, $\chi^2 = \sum_k \text{pull}_k^2$, and the sum of the squared pulls of the four spin averages, $\chi_{\overline{M}}^2$, is the part of the χ^2 that comes from the position of the multiplets as a whole. These pulls use only the lattice error: the model error σ_E of Table 5 is not included (with it, the pulls of the data points are the smaller ones quoted above).

Multiplet	J^{PC}	Lattice	No mix Best				Mix Best			
		$\overline{M}$ (GeV)	$\overline{M}$ (GeV)	Δ (MeV)	pull	χ_{pts}^2	$\overline{M}$ (GeV)	Δ (MeV)	pull	χ_{pts}^2
$(s/d)_1$	$1^{--}, 0^{-+}, 1^{-+}, 2^{-+}$	4.012(12)	4.052	+40	-3.4	11.9	4.060	+48	-4.1	21.8
p_1	$1^{++}, 0^{+-}, 1^{+-}, 2^{+-}$	4.146(12)	4.160	+14	-1.2	6.0	4.161	+15	-1.2	11.4
$(p/f)_2$	$2^{++}, 1^{+-}, 2^{+-}, 3^{+-}$	4.232(17)	4.279	+47	-2.9	8.6	4.282	+50	-3.0	13.1
p_0	$0^{++}, 1^{+-}$	4.495(42)	4.475	-20	+0.5	0.7	4.493	-2	+0.1	1.3
χ^2 of the 14 points			27.2				47.7			
$\chi_{\overline{M}}^2$ of the 4 spin averages			21.3				27.6			
χ^2 left if each multiplet is shifted freely			11.3 ($\chi^2/\text{dof} = 1.41$)				23.2 ($\chi^2/\text{dof} = 2.90$)			
χ^2 with the best common shift			13.2, -25 MeV ($\chi^2/\text{dof} = 1.20$)				26.2, -31 MeV ($\chi^2/\text{dof} = 2.38$)			

Table 6: Spin averages of the charmonium hybrid multiplets at the best fits of Table 5, from **spinAverages.m**. Lattice: $\overline{M}_{\text{lat}}(\sigma_{\overline{M}})$. $\Delta = \overline{M}_{\text{model}} - \overline{M}_{\text{lat}}$ and $\text{pull} = (\overline{M}_{\text{lat}} - \overline{M}_{\text{model}})/\sigma_{\overline{M}}$. χ_{pts}^2 is the contribution of the points of each multiplet to the χ^2 of the fit. The last two rows are at fixed A, B (no refit): each multiplet shifted by its own constant ($\text{dof} = 8$), or all levels shifted by the same constant ($\text{dof} = 11$), which is equivalent to $E_{\text{val}} = 0.475$ GeV (no mix) or 0.481 GeV (mix) instead of 0.45 GeV.

Table 6 shows that without mixing the splittings are well reproduced: shifting each multiplet freely leaves $\chi^2/\text{dof} = 1.41$, and a common shift of -25 MeV alone brings the χ^2 from 27.2 to 13.2. The tension is in the position of the $(s/d)_1$ and $(p/f)_2$ multiplets, 40–50 MeV above the lattice (-3.4σ and -2.9σ), while p_1 and p_0 agree. With mixing the $(s/d)_1$ and $(p/f)_2$ averages move further up and, in addition, the splittings are worse (23.2 left after the free shifts), mainly the $(p/f)_2$ 1^{+-} – 3^{+-} splitting (202 MeV in the model vs. 73 MeV on the lattice) and the p_1 1^{++} spin-0 state, 47 MeV above its spin average.

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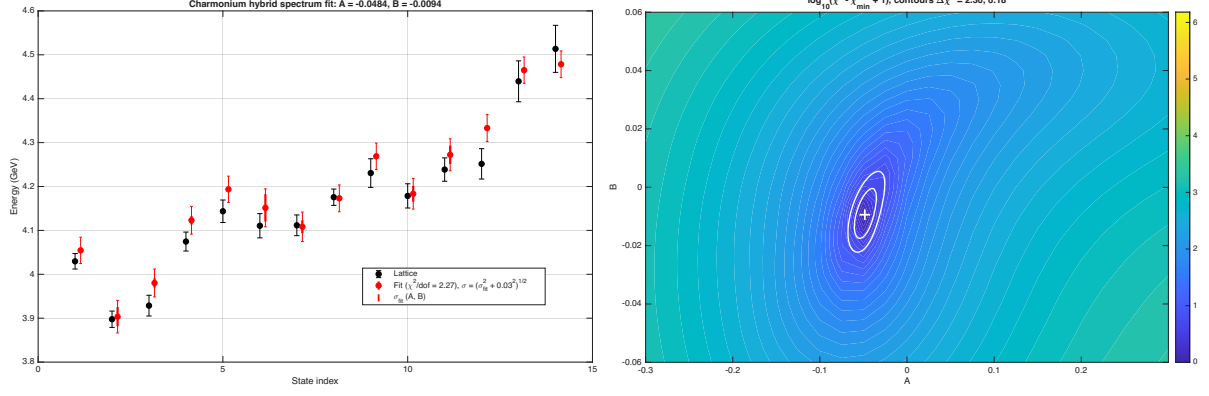


Figure 2: Best fit without mixing. Left: lattice data (black) and fitted energies (red, shifted to the right) with σ_E (thin bar) and σ_{fit} (thick bar), in the order of Table 5. Right: $\log_{10}(\chi^2 - \chi_{\min}^2 + 1)$ in the (A, B) plane with the contours $\Delta\chi^2 = 2.30$ and 6.18 (68% and 95% for two parameters, unscaled) around the minimum (cross). The tilt of the contours is the correlation $\rho(A, B)$.

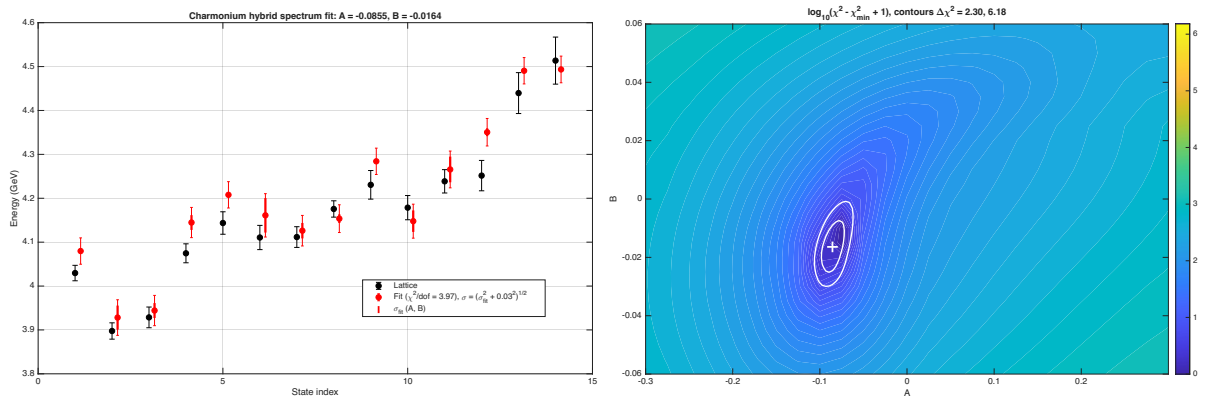


Figure 3: Same as Fig. 2 for the best fit with mixing.

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